

Module 3

Sinusoidal Steady-State Analysis

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Kirchhoff's Laws in Frequency domain

Let $v_1, v_2, \dots, v_n$ be the voltages around a closed loop.

$$v_1 + v_2 + \dots + v_n = 0$$

$$V_{m1} \cos(\omega t + \theta_1) + V_{m2} \cos(\omega t + \theta_2) \\ + \dots + V_{mn} \cos(\omega t + \theta_n) = 0$$

$$\text{Re}(V_{m1} e^{j\theta_1} e^{j\omega t}) + \text{Re}(V_{m2} e^{j\theta_2} e^{j\omega t}) + \dots + \text{Re}(V_{mn} e^{j\theta_n} e^{j\omega t}) = 0$$

$$\text{Re}[(V_{m1} e^{j\theta_1} + V_{m2} e^{j\theta_2} + \dots + V_{mn} e^{j\theta_n}) e^{j\omega t}] = 0$$

$$\text{Re}[(\mathbf{V}_1 + \mathbf{V}_2 + \dots + \mathbf{V}_n) e^{j\omega t}] = 0$$

Since $e^{j\omega t} \neq 0$,

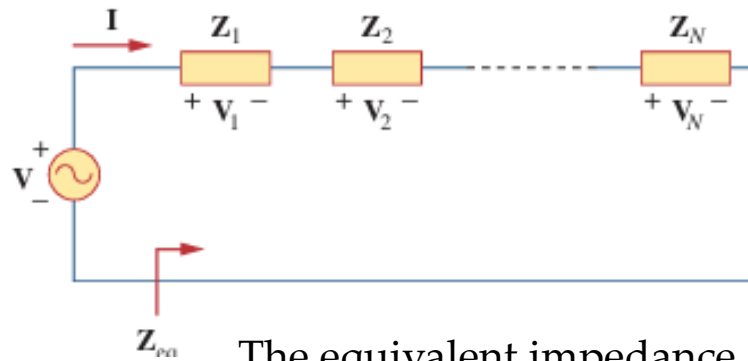
$$\mathbf{V}_1 + \mathbf{V}_2 + \dots + \mathbf{V}_n = 0$$

This indicates that Kirchhoff's voltage law holds for phasors.

Similarly,

$$\mathbf{I}_1 + \mathbf{I}_2 + \dots + \mathbf{I}_n = 0$$

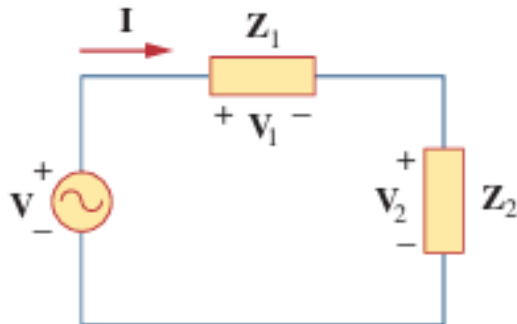
Impedance combinations



$$V = V_1 + V_2 + \dots + V_N = I(Z_1 + Z_2 + \dots + Z_N)$$

The equivalent impedance at the input terminals is, $Z_{eq} = \frac{V}{I} = Z_1 + Z_2 + \dots + Z_N$

$$Z_{eq} = Z_1 + Z_2 + \dots + Z_N$$

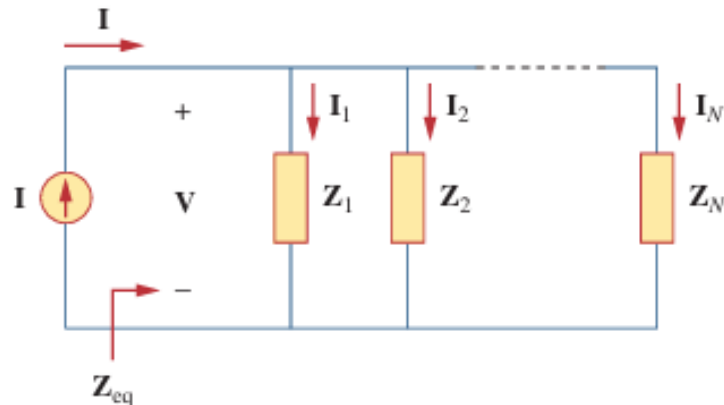


$$I = \frac{V}{Z_1 + Z_2}$$

$$V_1 = Z_1 I \text{ and } V_2 = Z_2 I$$

$$V_1 = \frac{Z_1}{Z_1 + Z_2} V, \quad V_2 = \frac{Z_2}{Z_1 + Z_2} V$$

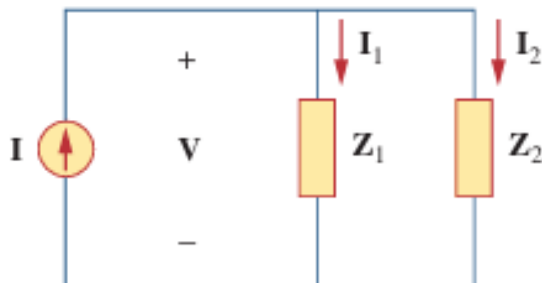
Kirchhoff's Laws in Frequency domain



$$I = I_1 + I_2 + \dots + I_N = V \left(\frac{1}{Z_1} + \frac{1}{Z_2} + \dots + \frac{1}{Z_N} \right)$$

$$\frac{1}{Z_{eq}} = \frac{I}{V} = \frac{1}{Z_1} + \frac{1}{Z_2} + \dots + \frac{1}{Z_N}$$

$$Y_{eq} = Y_1 + Y_2 + \dots + Y_N$$

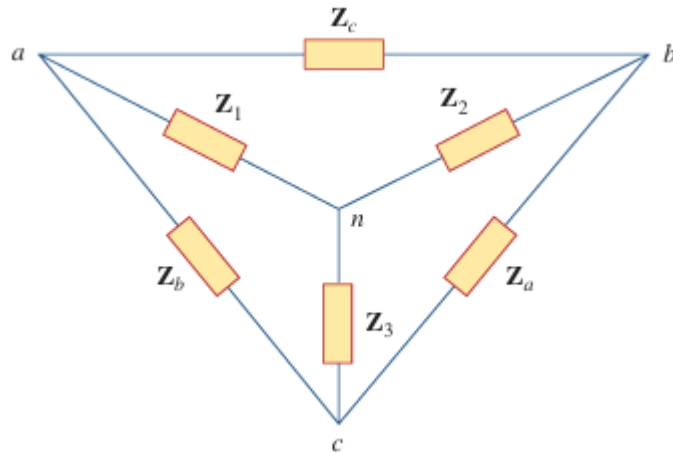


$$V = IZ_{eq} = I_1 Z_1 = I_2 Z_2$$

$$I_1 = \frac{Z_2}{Z_1 + Z_2} I, \quad I_2 = \frac{Z_1}{Z_1 + Z_2} I$$

Delta-Star transformations

The delta-to-wye and wye-to-delta transformations that we applied to resistive circuits are also valid for impedances.



Y- Δ Conversion:

$$\begin{aligned} Z_a &= \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_1} \\ Z_b &= \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_2} \\ Z_c &= \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_3} \end{aligned}$$

Δ -Y Conversion:

$$\begin{aligned} Z_1 &= \frac{Z_b Z_c}{Z_a + Z_b + Z_c} \\ Z_2 &= \frac{Z_c Z_a}{Z_a + Z_b + Z_c} \\ Z_3 &= \frac{Z_a Z_b}{Z_a + Z_b + Z_c} \end{aligned}$$

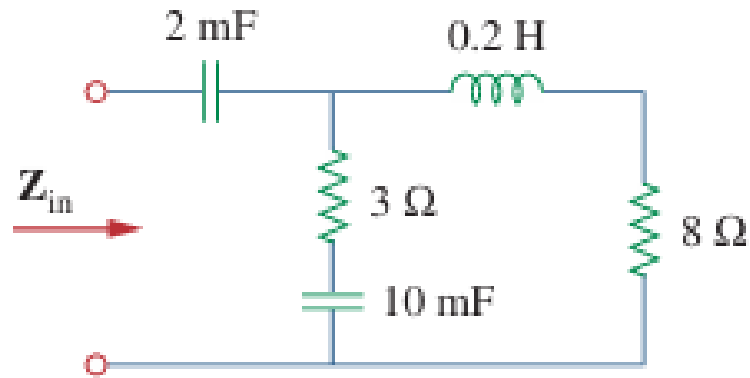
A delta or wye circuit is said to be balanced if it has equal impedances in all three branches.

$$Z_{\Delta} = 3Z_Y \quad \text{or} \quad Z_Y = \frac{1}{3} Z_{\Delta}$$

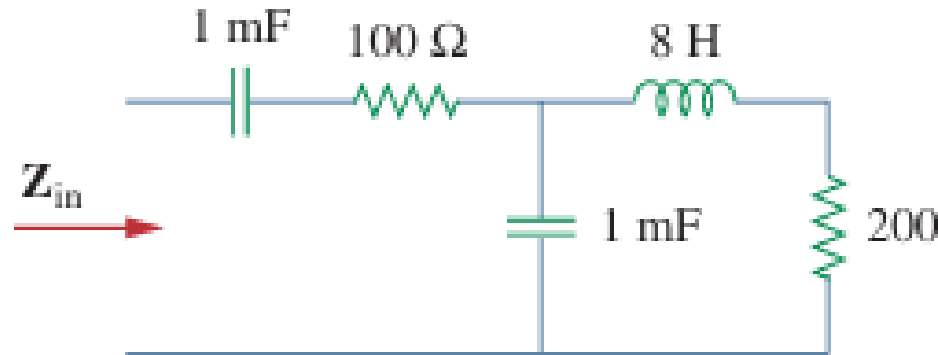
where $Z_Y = Z_1 = Z_2 = Z_3$ and $Z_{\Delta} = Z_a = Z_b = Z_c$.

Exercise

1. Find the input impedance of the circuit. Assume that the circuit operates at $\omega = 50$ rad/s.



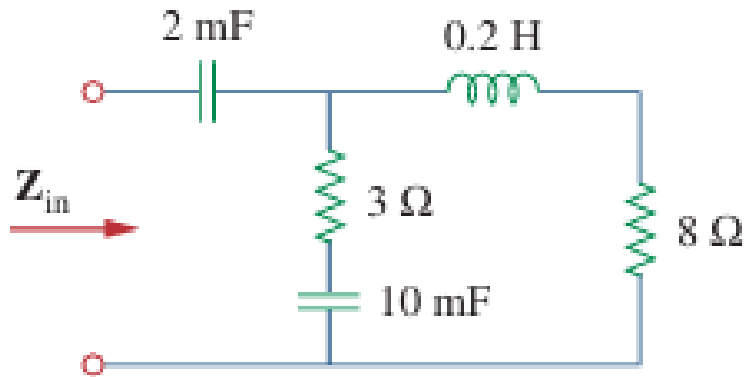
2. Find the input impedance of the circuit. Assume that the circuit operates at $\omega = 10$ rad/s.



$$Z_{eq} = 149.52 - j195$$

Exercise

- Find the input impedance of the circuit. Assume that the circuit operates at $\omega = 50$ rad/s.



$$Z_1 = \frac{1}{j\omega C} = \frac{1}{j50 \times 2 \times 10^{-3}} = -j10 \Omega$$

$$Z_2 = 3 + \frac{1}{j\omega C} = 3 + \frac{1}{j50 \times 10 \times 10^{-3}} = (3 - j2) \Omega$$

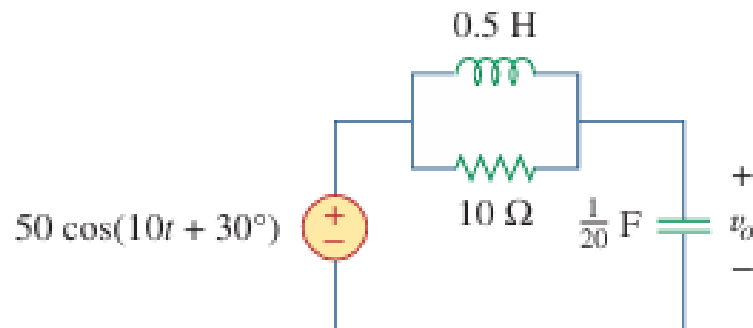
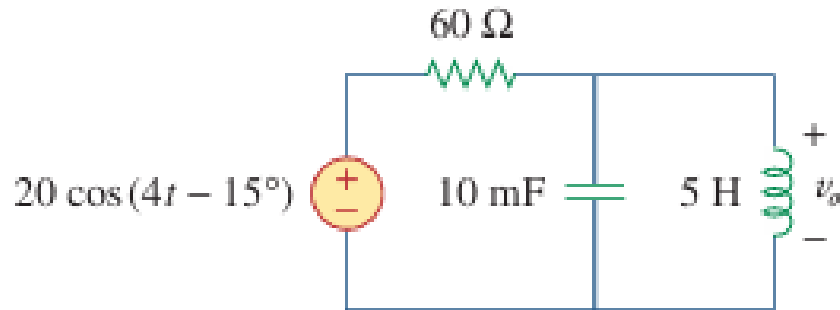
$$Z_3 = 8 + j\omega L = 8 + j50 \times 0.2 = (8 + j10) \Omega$$

$$\begin{aligned} Z_{\text{in}} &= Z_1 + Z_2 \parallel Z_3 = -j10 + \frac{(3 - j2)(8 + j10)}{11 + j8} \\ &= -j10 + \frac{(44 + j14)(11 - j8)}{11^2 + 8^2} = -j10 + 3.22 - j1.07 \Omega \end{aligned}$$

$$Z_{\text{in}} = 3.22 - j11.07 \Omega$$

Exercise

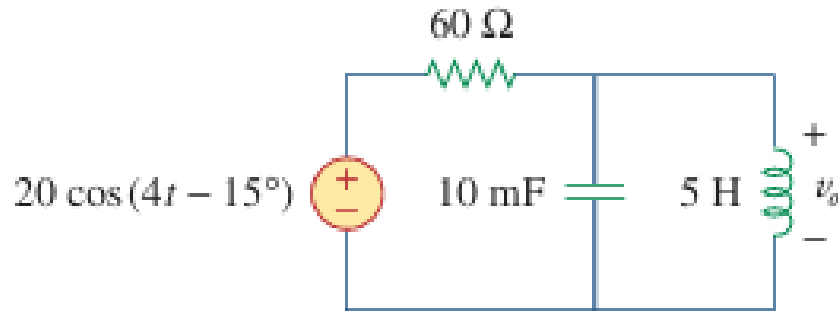
3. Determine $v_o(t)$ in both the circuits.



$$\text{Ans: } v_o(t) = 35.36 \cos(10t - 105^\circ) \, \text{V}$$

Exercise

3. Determine $v_o(t)$ in both the circuits.



$$v_s = 20 \cos(4t - 15^\circ) \Rightarrow \mathbf{V}_s = 20 \angle -15^\circ \text{ V}, \quad \omega = 4$$

$$10 \text{ mF} \Rightarrow \frac{1}{j\omega C} = \frac{1}{j4 \times 10 \times 10^{-3}} = -j25 \Omega$$

$$5 \text{ H} \Rightarrow j\omega L = j4 \times 5 = j20 \Omega$$

Then $\mathbf{Z}_1 = 60 \Omega$ and

$$\mathbf{Z}_2 = -j25 \parallel j20 = \frac{-j25 \times j20}{-j25 + j20} = j100 \Omega$$

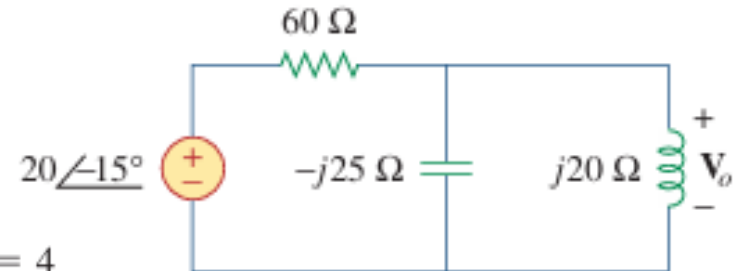
By the voltage-division principle,

$$\begin{aligned} \mathbf{V}_o &= \frac{\mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2} \mathbf{V}_s = \frac{j100}{60 + j100} (20 \angle -15^\circ) \\ &= (0.8575 \angle 30.96^\circ) (20 \angle -15^\circ) = 17.15 \angle 15.96^\circ \text{ V} \end{aligned}$$

We convert this to the time domain and obtain

$$v_o(t) = 17.15 \cos(4t + 15.96^\circ) \text{ V}$$

phasor domain equivalent

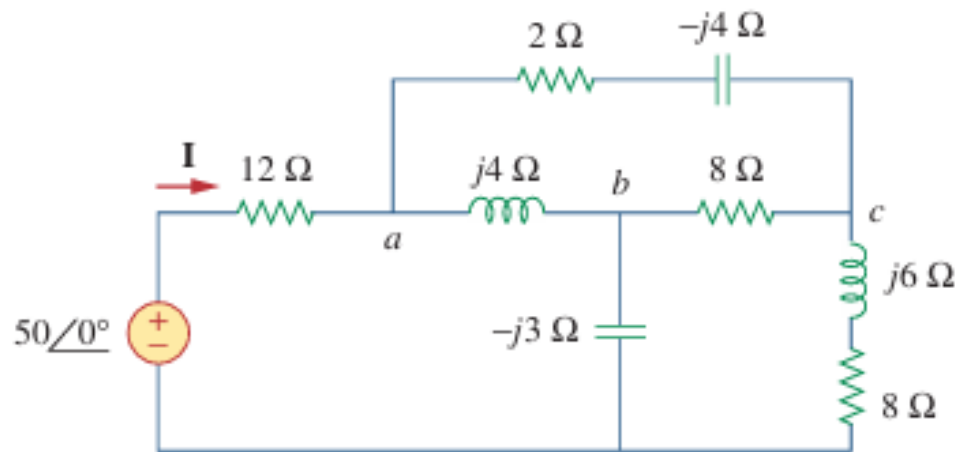


$\mathbf{Z}_1$ = Impedance of the 60- Ω resistor

$\mathbf{Z}_2$ = Impedance of the parallel combination of the 10-mF capacitor and the 5-H inductor

Exercise

4. Find current I in the circuit.

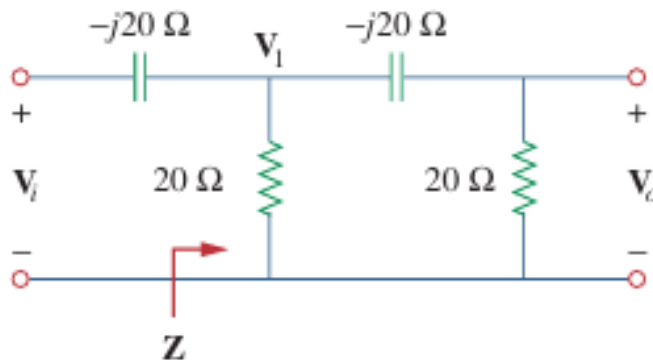


Exercise

5. Design an RC circuit to provide a phase of 90° leading.

$$\tan^{-1} X_C/R = 45^\circ \text{ means, } R = X_C.$$

By cascading two similar RC circuits, we can design a circuit to introduce a phase shift positive or leading phase of 90° . So, assume $R = \text{Mod}X_C = 20 \Omega$



$$\mathbf{Z} = 20 \parallel (20 - j20) = \frac{20(20 - j20)}{40 - j20} = 12 - j4 \Omega$$

$$\mathbf{V}_1 = \frac{\mathbf{Z}}{\mathbf{Z} - j20} \mathbf{V}_i = \frac{12 - j4}{12 - j24} \mathbf{V}_i = \frac{\sqrt{2}}{3} \angle 45^\circ \mathbf{V}_i$$

$$\mathbf{V}_o = \frac{20}{20 - j20} \mathbf{V}_1 = \frac{\sqrt{2}}{2} \angle 45^\circ \mathbf{V}_1$$

$$\mathbf{V}_o = \left(\frac{\sqrt{2}}{2} \angle 45^\circ \right) \left(\frac{\sqrt{2}}{3} \angle 45^\circ \mathbf{V}_i \right) = \frac{1}{3} \angle 90^\circ \mathbf{V}_i$$

Next class: Kirchhoff's laws in frequency domain;
instantaneous and average power; complex power